

Differentially Private Steering for Large Language Model Alignment



The hypothesis is clear, novel, and addresses a meaningful problem: achieving LLM alignment via activation editing while maintaining Differential Privacy (DP) for the demonstration data. The experimental design is highly rigorous and comprehensive, addressing all critical aspects of the hypothesis. Experiment 1 evaluates the core alignment efficacy across multiple modern LLM families (Llama-2, Mistral, Gemma, Qwen) and established safety/alignment benchmarks, using appropriate non-private and zero-shot baselines. Experiment 2 and 3 thoroughly assess the 'utility' side of the privacy-utility tradeoff by measuring open-ended generation quality (via GPT-4 evaluation) and general capability retention (via MMLU), ensuring the steering method does not degrade the model's core competencies. Experiment 4 provides valuable insights into scaling behaviors across model sizes. Finally, Experiment 5 is a standout addition, proposing a novel Membership Inference Attack (MIA) to empirically audit the theoretical DP guarantees, **which is a gold standard in privacy research. The metrics, baselines, and ablations are well-chosen and perfectly aligned with the claims being tested.**